

AGGREGATE TRENDS PRESENTATION 3

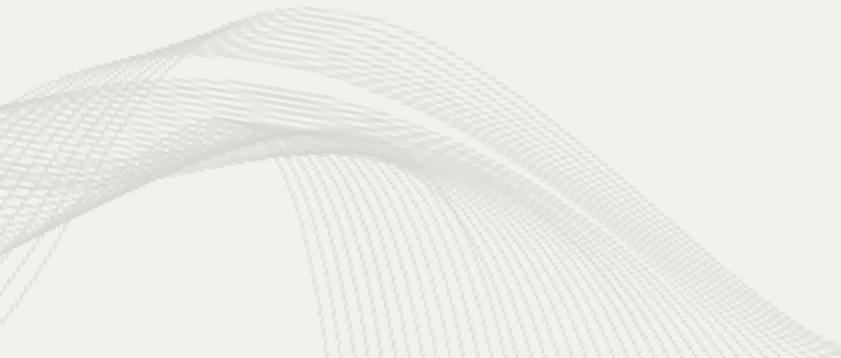
Prepared by: Edwin, Ella, Liv & Tatum

Wednesday, November 5th



Redirection Analysis

Understanding how and why calls are redirected for the top called numbers.



8300 is the most frequently occurring called number in the data.

Out of 516786 unique call sessions, 117579 (22.8%) ended with 8300 as the called number in the call's last leg...

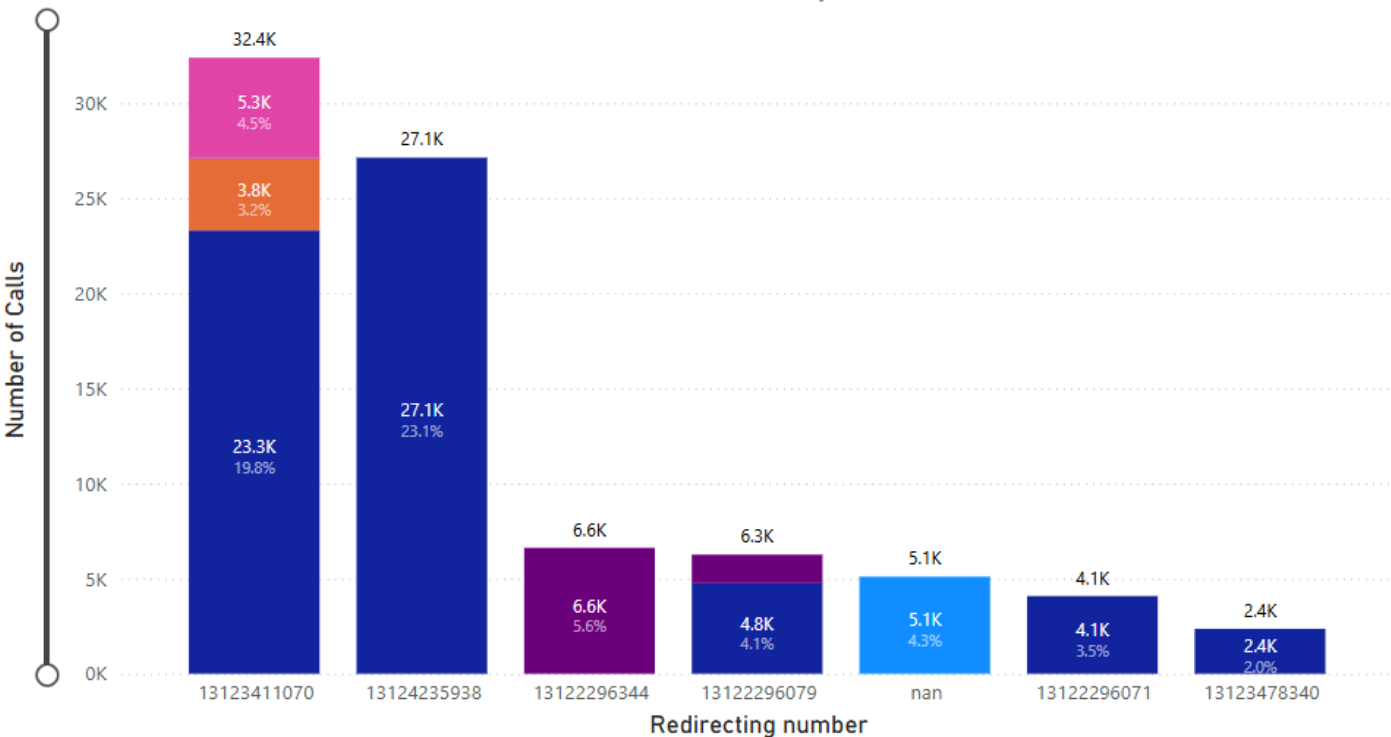
1. What numbers are redirecting the calls to 8300 at the end?
2. Why are those numbers redirecting to 8300?

Looking at the numbers that redirect to 8300 at the end of calls

Distribution of Redirecting Numbers at the Ends of Calls Ending With 8300

Percentages are based on the proportion of ALL calls that end in 8300 (e.g., 1070 redirecting to 8300 because of NoAnswer is 19.8% of all calls ending in 8300).

Redirect reason nan NoAnswer Unavailable Unconditional UserBusy



Range of columns to show

Min rank - max rank

1

7



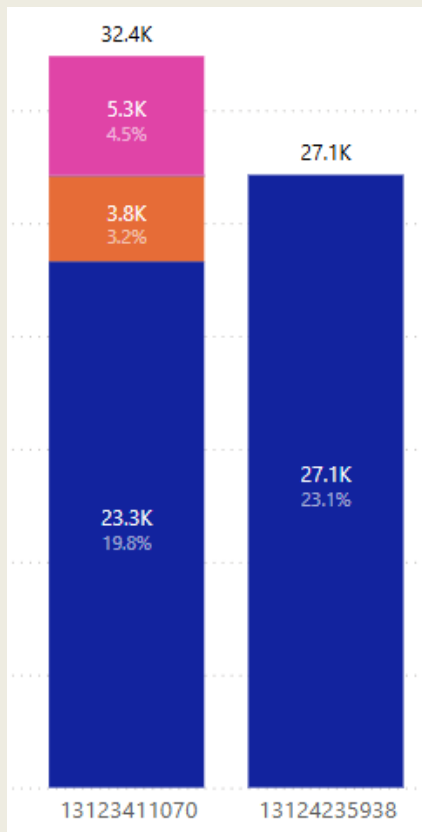
Phone Number Descriptions

- 3411070 - Main number
- 4235938 - legalclinics (Community Legal Clinics)
- 2296344 - Bankruptcy Helpdesk VM
- 2296079 - Nursing Home Ombudsman
- 2296071 - Criminal Records
- 3478340 - Veterans Rights Project VM
- 4235904 - Austin Intake VM
- 2296072 - JEHD (Juvenile Expungement Help Desk)
- 2296300 - Direct Line to Front Desk?
- 2296348 - (no description given)

Looking at the numbers that redirect to 8300 at the end of calls

Analysis

Redirect reason ● nan ● NoAnswer ● Unavailable ● Unconditional ● UserBusy



Insights:

- 1070 (Main number) accounted for a total of 27.5% of the redirections to 8300
- About 72% of those redirections were because of NoAnswer.
- The remaining 28% of those redirections were either because of Unavailable or UserBusy.

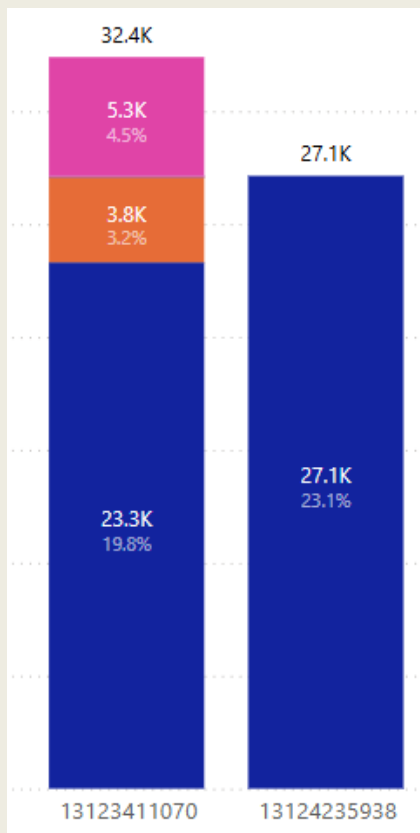
Altogether:

- 1070 appears to redirect to 8300 for the purpose of voicemail.
- These calls that end in 1070 redirecting to 8300 are due to 1070 not being available.

Looking at the numbers that redirect to 8300 at the end of calls

Analysis

Redirect reason ● nan ● NoAnswer ● Unavailable ● Unconditional ● UserBusy



Insights:

- 5938 (Community legalclinics) accounted for a total of 23.1% of the redirections to 8300
- All redirections were due to NoAnswer

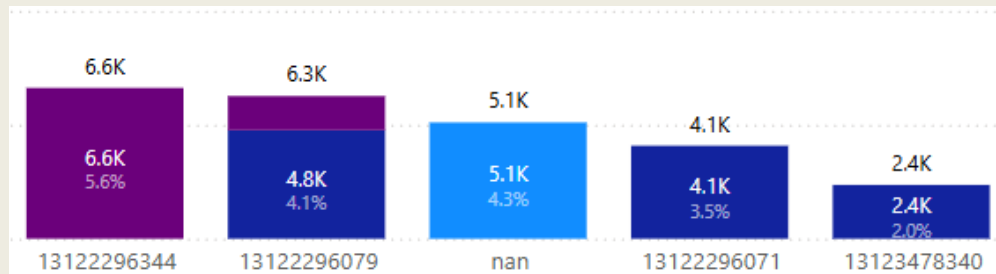
Altogether:

- 8300 also serves as the voicemail line after 5938 does ends up being unavailable

Looking at the numbers that redirect to 8300 at the end of calls

Analysis

Redirect reason ● nan ● NoAnswer ● Unavailable ● Unconditional ● UserBusy



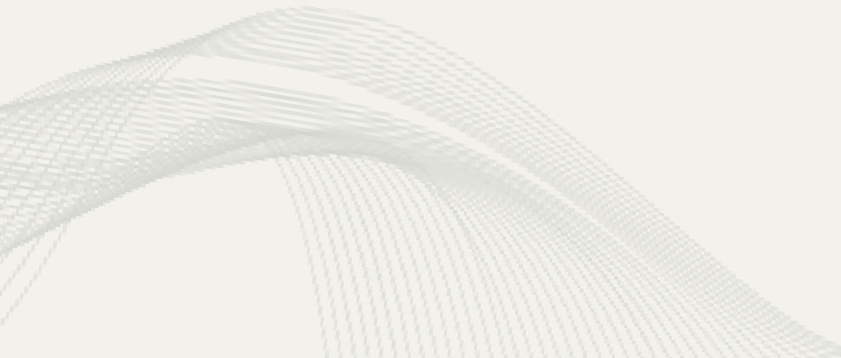
Insights:

- The remaining most common scenarios all show similar results.
- Altogether, this confirms that **8300 simply serves as the voicemail line that numbers redirect to when they are not busy**. The 8300 line being at the end of a call likely indicates an unsuccessful call.
- There are no actionable recommendations from my analysis at this time.

Next steps

1. For each redirecting number, how often do they result in calls ending in 8300 (voicemail)?
2. Looking further back into the call journey
3. Taking other columns into consideration

Front Desk Analysis



Only a small share of calls reach the Front Desk directly.

Distribution of Front Desk Call Activity

2488

Count of Correlation ID

Redirecting number

- ☒ (Blank)
☐ 13123411070
☐ 13125068646
☐ 13125068647

Direction

- ☐ ORIGINATING
☒ TERMINATING

Distribution of Front Desk Call Activity

199

Count of Correlation ID

Redirecting number

- ☒ (Blank)
☐ 13123411070
☐ 13125068646
☐ 13125068647

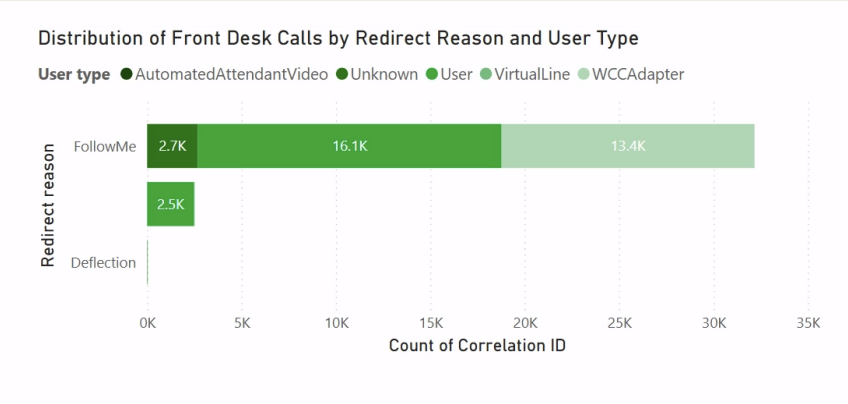
Direction

- ☒ ORIGINATING
☐ TERMINATING

What We Found:

- Of all calls involving the Front Desk, only about ~ **13%** are placed directly to that number
- These calls likely represent individuals who already have appointments

Most Front Desk Calls are redirected from the Main Line.



What We Found:

- The **majority of Front Desk calls (~15.9k)** are redirected from the Main Line (1070)
- These calls are routed through the system using the **“Follow Me” redirect reason**, which sends unanswered main line calls to the Front Desk
- Nearly **half of redirected calls involve system routing (“WCC Adaptor”)** showing that many front desk calls are part of automated handling

When calls continue past the Front Desk, most route back to the Main line.

Top immediate next hops after FD (in inbound journeys where FD is not last):

TERMINATING	to:13123411070 from:13123411070	3532
ORIGINATING	to:13122296300 from:13123411070	1070
TERMINATING	to:13125068646 from:13125068646	404
ORIGINATING	to:13123478300 from:13124235938	393
TERMINATING	to:13123478300 from:13123478300	298
TERMINATING	to:1180 from:1180	287
TERMINATING	to:1182 from:1182	140
TERMINATING	to:13125068647 from:13125068647	107
TERMINATING	to:13122296346 from:13122296346	27
TERMINATING	to:13123478342 from:13123478342	27
ORIGINATING	to:13123478300 from:13122296073	19
TERMINATING	to:13123478375 from:13123478375	14
ORIGINATING	to:13125068649 from:13124235938	13
ORIGINATING	to:13123478300 from:13122296346	12
ORIGINATING	to:2302 from:13123411070	12

* See Appendix A for supporting Python code chunk used to generate this analysis

What We Found:

- About **3,500** calls *return* to the Main Line after reaching the Front Desk
- Around **1,000** of those are *sent back* to the Front Desk again within that same call journey
- This suggests callers may not always be able to navigate the automated menu and **rely on the person at the Front Desk** for help

Roughly 2/3 of inbound calls end at the Front Desk.



■ FD is the final leg ■ Call journey does not end at FD

Unique journeys with an FD inbound leg: 18,591
FD is the final leg in inbound journeys: 11,891/18,591 (63.96%).

** See Appendix B for supporting Python code chunk used to generate this analysis*

What We Found:

- Out of the call journeys that reached the front desk, **~64% ended there**
- The remaining **~36% were transferred elsewhere** (main line or internal extensions)
- Most callers' needs **are resolved at the Front Desk** but a substantial minority still require additional routing

Recommendation

1

Simplify call routing and empower Front Desk resolution.

Some callers may be struggling to reach their destination through the automated menu and rely on Front Desk staff for guidance.

- Identify which types of callers or menu paths lead to confusion
- Determine how to make it easier for callers to get to the right place without needing to rely on the Front Desk as a middle step



Next Steps: What the Final Front Desk Dashboard Would Include

1

Call Volume Trends

Line chart showing Front Desk call volume over time

2

Call Flow Map

Shows % of calls that end at Front Desk vs. redirect elsewhere

3

Journey Outcomes

% of calls ending at FD, % returning to Main Line, % transferred to queues, % directly dialed to Front Desk

4

Caller Path Analysis

Bar chart with op redirecting numbers to FD & Table of Top Next Hops After Front Desk

5

Performance Metrics

KPIs for average call duration, ring time and answer rate at FD

6

Integration With CAR Data

This will link FD transfers to specific queues and enable more insight into why callers are redirected

Submenu Analysis

Hierarchical Menu Usage Analysis: Power BI Drill-Down Dashboard



Preparing the Data for Drill-Down Analysis

1	Contact Session ID	EP Name	Flow Name	Activity Name	Activity Start Time	hour
2	001a3748-8d50-4550-8461-33547983de0	Main Number Telephony EP			1/14/25 12:39	12
3	001a3748-8d50-4550-8461-33547983de0		LACMain		1/14/25 12:39	12
4	001a3748-8d50-4550-8461-33547983de0	Main Number Telephony EP		LanguageSelectionMen	1/14/25 12:39	12
5	001a3748-8d50-4550-8461-33547983de0	Main Number Telephony EP	LACMain		1/14/25 12:39	12
6	001a3748-8d50-4550-8461-33547983de0	Main Number Telephony EP	MainMenu		1/14/25 12:39	12
7	001a3748-8d50-4550-8461-33547983de0		PreLegalMenuSeniorsMenu		1/14/25 12:40	12
8	001a3748-8d50-4550-8461-33547983de0	Pre-Legal Menu Seniors Menu	Telephony EP	SeniorsMenu	1/14/25 12:40	12
9	001a3748-8d50-4550-8461-33547983de0		LegalMenu		1/14/25 12:40	12
10	001a3748-8d50-4550-8461-33547983de0	Legal Menu Telephony EP		LegalMenu1	1/14/25 12:40	12
11	001a3748-8d50-4550-8461-33547983de0	Legal Menu Telephony EP		LegalMenu2	1/14/25 12:40	12
12	001a3748-8d50-4550-8461-33547983de0		LegalEmploymentMenu		1/14/25 12:43	12
13	001a3748-8d50-4550-8461-33547983de0	Legal Employment Menu Telephony EP		EmploymentMenu	1/14/25 12:43	12
14	001a3748-8d50-4550-8461-33547983de0		Queues		1/14/25 12:43	12
15	001a3748-8d50-4550-8461-33547983de0		ClosedQueueMenu		1/14/25 12:43	12
16	001a3748-8d50-4550-8461-33547983de0	Closed Queue Menu Telephony EP		ClosedQueueMenu	1/14/25 12:43	12
17	001a3748-8d50-4550-8461-33547983de0	Closed Queue Menu Telephony EP		ClosedQueueMenu	1/14/25 12:44	12
18	001a3748-8d50-4550-8461-33547983de0	Closed Queue Menu Telephony EP			1/14/25 12:44	12
19	00229391-8614-4eb1-b1e0-e0a78076e0e4	Main Number Telephony EP			1/14/25 14:50	14
20	00229391-8614-4eb1-b1e0-e0a78076e0e4		LACMain		1/14/25 14:50	14
21	00229391-8614-4eb1-b1e0-e0a78076e0e4	Main Number Telephony EP		LanguageSelectionMen	1/14/25 14:50	14
22	00229391-8614-4eb1-b1e0-e0a78076e0e4	Main Number Telephony EP	LACMain		1/14/25 14:50	14
23	00229391-8614-4eb1-b1e0-e0a78076e0e4	Main Number Telephony EP	MainMenu		1/14/25 14:50	14
24	00229391-8614-4eb1-b1e0-e0a78076e0e4		PreLegalMenuSeniorsMenu		1/14/25 14:50	14
25	00229391-8614-4eb1-b1e0-e0a78076e0e4	Pre-Legal Menu Seniors Menu	Telephony EP	SeniorsMenu	1/14/25 14:50	14
26	00229391-8614-4eb1-b1e0-e0a78076e0e4		LegalMenu		1/14/25 14:50	14
27	00229391-8614-4eb1-b1e0-e0a78076e0e4	Legal Menu Telephony EP		LegalMenu1	1/14/25 14:50	14
28	00229391-8614-4eb1-b1e0-e0a78076e0e4	Legal Menu Telephony EP		LegalMenu2	1/14/25 14:50	14
29	00229391-8614-4eb1-b1e0-e0a78076e0e4		LegalFamilyMenu		1/14/25 14:52	14
30	00229391-8614-4eb1-b1e0-e0a78076e0e4	Legal Family Menu Telephony EP		FamilyMenu	1/14/25 14:52	14
31	00229391-8614-4eb1-b1e0-e0a78076e0e4		Queues		1/14/25 14:52	14
32	00229391-8614-4eb1-b1e0-e0a78076e0e4		ClosedQueueMenu		1/14/25 14:52	14
33	00229391-8614-4eb1-b1e0-e0a78076e0e4	Closed Queue Menu Telephony EP		ClosedQueueMenu	1/14/25 14:52	14
34	00229391-8614-4eb1-b1e0-e0a78076e0e4	Closed Queue Menu Telephony EP			1/14/25 14:53	14
35	00229391-8614-4eb1-b1e0-e0a78076e0e4	Closed Queue Menu Telephony EP		ClosedQueueMenu	1/14/25 14:53	14
36	0020e73-cd5-4274-304e-ee8a2c7088b	Main Number Telephony EP			1/13/25 10:54	10

1	Contact Sess Date	hour	Current Activ	Second Activ	Third Activity	Fourth Activ	Fifth Activity
2	00014a58-a6	2/28/25	8 Family	Closed Queue	Abandoned /	Abandoned /	Abandoned / End of call
3	00047fa4-ea	7/26/24	11 Family	Closed Queue	Closed Queue	Closed Queue	DisconnectContact1
4	0005118f-8d	10/23/24	14 Family	Divorce / Par	Closed Queue	Closed Queue	Closed QueueMenu
5	0005118df-20	9/4/24	15 Family	Closed Queue	Closed Queue	Abandoned /	Abandoned / End of call
6	0006a469-2f	12/10/24	10 Family	Closed Queue	Abandoned /	Abandoned /	Abandoned / End of call
7	0007108a-5c	3/26/25	16 Family	Divorce / Par	Closed Queue	Closed Queue	ClinicVoicemailTransfer
8	0008a3cb-6e	10/21/24	15 Family	Divorce / Par	Closed Queue	Closed Queue	Abandoned / End of call
9	000a9e4a-e5	6/26/25	15 Family	Divorce / Par	Simple Divor	ClinicVoicen	Abandoned / End of call
10	000bbfa2-0e	9/25/24	15 Family	Divorce / Par	Child Suppor	DivorceOrPa	Abandoned / End of call
11	000e04be-c5	5/20/25	12 Family	Closed Queue	Abandoned /	Abandoned /	Abandoned / End of call
12	001761ac-8c	4/29/25	9 Family	Closed Queue	Closed Queue	Abandoned /	Abandoned / End of call
13	001c0d31-8c	3/27/25	16 Family	Divorce / Par	Closed Queue	Closed Queue	Abandoned / End of call
14	001c6390-a0	2/24/25	11 Family	Divorce / Par	Child Suppor	DivorceOrPa	Abandoned / End of call
15	001dc39d-2a	4/18/25	10 Family	Family Queue	IntakePreQue	FamilySPQue	PreQueueMessage2
16	00216490-b7	7/11/24	9 Family	Divorce / Par	Simple Divor	Closed Queue	Closed QueueMenu
17	00223fe9-e1	2/26/25	9 Family	Closed Queue	Closed Queue	Abandoned /	Abandoned / End of call
18	002272e2-5f	9/24/25	11 Family	Divorce / Par	Simple Divor	ClinicVoicen	Abandoned / End of call
19	00229391-8e	1/14/25	14 Family	Closed Queue	Closed Queue	Abandoned /	Abandoned / End of call
20	0022a2ee-f1	8/7/24	13 Family	Closed Queue	Closed Queue	Abandoned /	Abandoned / End of call
21	002309ae-f7	6/26/25	12 Family	Divorce / Par	Closed Queue	Closed Queue	Abandoned / End of call
22	0024ac55-5c	8/20/25	12 Family	Closed Queue	Closed Queue	ClinicVoicen	Abandoned / End of call
23	00274462-dc	2/24/25	13 Family	Divorce / Par	Child Suppor	DivorceOrPa	Abandoned / End of call
24	00278c5c-3f	11/19/24	10 Family	LegalMenu2	FrontDeskTrs	Abandoned /	Abandoned / End of call
25	00281d69-e5	5/30/25	9 Family	Closed Queue	Abandoned /	Abandoned /	Abandoned / End of call
26	002bfe03-2d	4/1/25	10 Family	Closed Queue	Abandoned /	Abandoned /	Abandoned / End of call
27	002bfd4f-e65	11/4/24	10 Family	Closed Queue	Abandoned /	Abandoned /	Abandoned / End of call

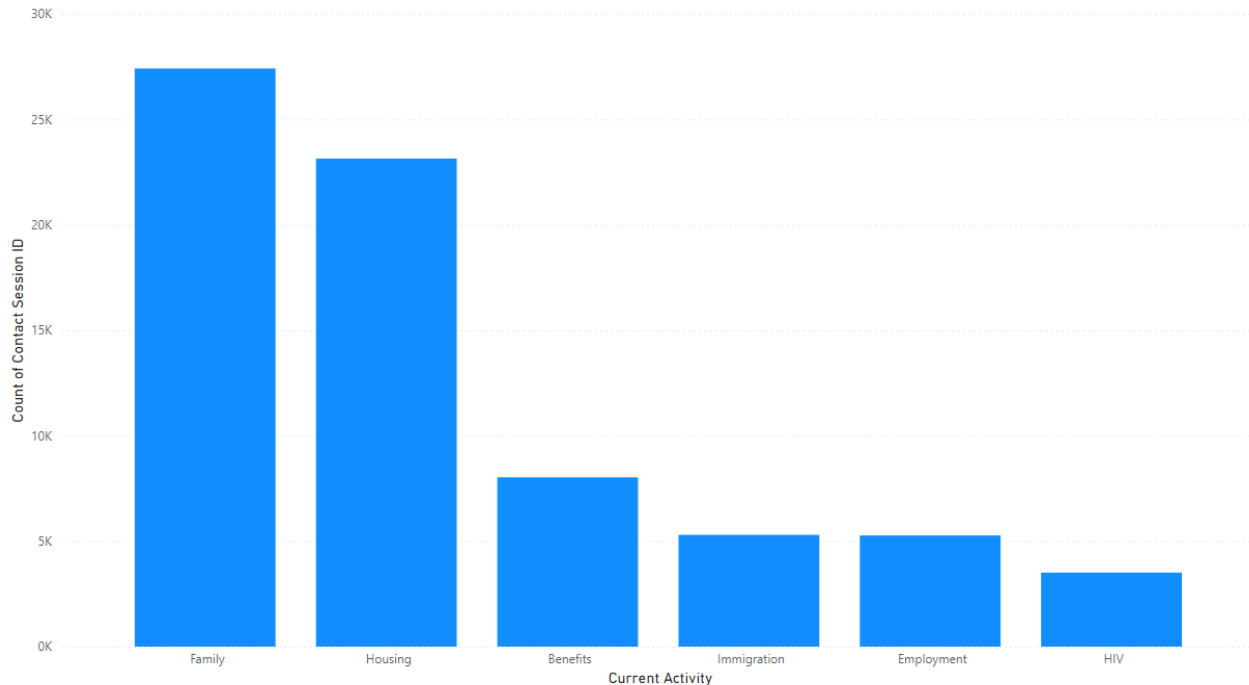
What We Did:

- Altered pre-existing code with Activities 1-3 to also display Activities 4-5
- Expanded the data to include activity choices from the Pre-Legal Menu

Activity Drill Downs for Legal and Pre-Legal Seniors Menus

Count of Contact Session ID

BY CURRENT ACTIVITY



What We Found:

- “Closed Queue” appears surprisingly high, nearly as common as Divorce/Parenting, which suggests many calls are reaching a non-live endpoint.
- This means even after navigating multiple menus, callers are often being routed to **Closed Queue** instead of connecting to an agent or resolution menu.

Recommendations

1

Update menu routing to prevent callers from reaching Closed Queue when lines are closed:

- Redirect callers to a general information message or callback option instead of an inactive queue.
- This reduces frustration and repeat calls while ensuring callers still receive guidance or next steps.



2

Extend queue hours or use smart overflow routing to redirect callers to available agents in related departments:

- Implement flexible routing so callers in high-volume areas like **Family** or **Housing** are automatically connected to open or related lines when their primary queue is closed.
- This improves live-answer rates and minimizes call abandonment during peak periods.



Next Steps & Further Questions

Next Steps

- Apply the Activity-depth code (Activities 4 & 5) across all menus so each EP and Flow has consistent depth for drill-down analysis
- Explore options to adjust menu routing that would reduce the number of calls ending in Closed Queue
- Update the Power BI dashboard to reflect the expanded activity hierarchy and visualize call paths across all menus

Further Questions

1. Can Closed Queue calls be redirected or logged for follow-up automatically when queues are inactive?
2. How can we add the average call per day feature on Power BI?

Appendix

** Includes additional details and visuals from our analysis that were not included in the live presentation due to time constraints.*

Appendix A

The following data analysis was completed in **Python** for the **All Calls Dataset**. Tatum's full code script is [here](#).

```
fd_inbound = df[(df["called_number"] == FD) & (df["direction"] == "TERMINATING")]

# Distinct journeys that ever hit FD inbound
fd_inbound_journeys = fd_inbound["correlation_id"].dropna().unique()
n_fd_inbound_journeys = len(fd_inbound_journeys)
print(f"Unique journeys with an FD inbound leg: {n_fd_inbound_journeys:,}")

# Among inbound journeys, check if the last leg is an FD leg
fd_inbound_last = (
    df[df["correlation_id"].isin(fd_inbound_journeys)]
    .groupby("correlation_id", as_index=False)
    .apply(lambda g: ((g["is_last_leg"]) & (g["called_number"] == FD)).any())
)

# Clean up the result
fd_inbound_last = fd_inbound_last.rename(columns={None: "fd_is_last"})
fd_inbound_last_count = fd_inbound_last["fd_is_last"].sum()
pct_fd_inbound_last = (fd_inbound_last_count / n_fd_inbound_journeys * 100) if n_fd_inbound_journeys else 0

print(f"FD is the final leg in inbound journeys: {fd_inbound_last_count:,}/{n_fd_inbound_journeys:,} "
      f"({pct_fd_inbound_last:.2f}%).")
```

Appendix B

The following data analysis was completed in **Python** for the **All Calls Dataset**. Tatum's full code script is [here](#).

```
def next_hop_after_last_fd(g):
    fd_rows = g[(g["called_number"] == FD) & (g["direction"] == "TERMINATING")]
    if fd_rows.empty:
        return None
    last_fd_idx = fd_rows["leg_idx"].max()
    candidate = g[g["leg_idx"] == last_fd_idx + 1]
    if candidate.empty:
        return None
    r = candidate.iloc[0]
    return f'{{r.get("direction","")}}|to:{{r.get("called_number","")}}|from:{{r.get("user_number","")}}'

nonfinal_inbound = (
    df[df["correlation_id"].isin(fd_inbound_journeys)]
      .groupby("correlation_id", group_keys=False)
      .apply(lambda g: (g["is_last_leg"] & (g["called_number"] == FD)).any())
)

nonfinal_ids = set(fd_inbound_journeys) - set(nonfinal_inbound[nonfinal_inbound].index)
nhops = (
    df[df["correlation_id"].isin(nonfinal_ids)]
      .groupby("correlation_id", group_keys=False)
      .apply(next_hop_after_last_fd)
      .dropna()
)

print("\nTop immediate next hops after FD (in inbound journeys where FD is not last):")
print(nhops.value_counts().head(15))
```